Electro-Thermal HVDC Gas-Insulated Line (GIL) Simulations

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Field Grading Material (FGM) is a technique used to control the electric field stress in AC and DC under high voltage applications. FGM is performed by spatial distribution of electric conductivity within the HVDC-GIL spacer to reduce the electric field stress at the junction of SF_6 gas, spacer and conductor/enclosure and is investigated by DC-steady state in stationary simulations. However, in this work, also DC-on process is investigated. Under DC-on process condition, spatial distribution of electric conductivity model is not sufficient to control the electric field stress, since also transient electric field occurs during the process. Hence, combination of electric conductivity and permittivity spatial distribution models has to be applied to obtain a reduced electric field stress around the spacer, which is investigated by transient simulations.

***Index Terms*—electric conductivity, field grading material (FGM), Gas-Insulated Line (GIL), permittivity.**

I. Introduction

The requirement for the electrical energy is increasing day-by-day around the globe, which demands to transport the electrical energy efficiently. The transition to green energy causes a need to transmit the energy over long distances (e.g. off-shore wind parks in the North Sea to industry in the south of Germany). Energy can be transmitted through High Voltage Direct Current (HVDC) and High Voltage Alternating Current (HVAC). Due to the ohmic resistance, losses occur in AC or DC power supply. In AC power supply, there are additional losses (inductive loss, capacitive loss, radiation loss, skin-effect) that occur with the ohmic losses, which leads to high reactive power losses in AC compared with DC. Hence, transmission of energy for long distance is performed by HVDC. Gas insulated transmission lines are a suitable transmission technology for the incoming challenges, since it has advantages in reliability, space efficiency and voltage capacity. Different field control techniques with an application of conductivity models for the insulation materials within HVDC GIL is used to determine the electric field and to obtain reduced electric field stress. Under DC condition, the electric conductivity is decisive for the electric field distribution which depends on temperature. The electric conductivity model is not sufficient to control the electric field stress, as transient capacitive stress also occurs during the process. Hence, electro-thermal transient simulations need to be performed with combination of electric conductivity and permittivity [5] models has to be applied to obtain a reduced electric field stress around the spacer [1, 3].

II. GIL

In Fig. 1, the metallic enclosure of GIL has an angular space that has been filled with an insulating gas, such as sulphur hexafluoride (SF_6), while the central position of the high-voltage conductor is secured by a support insulator. The present work involves the simulation of a 2D-axisymmetric geometry of HVDC GIL, where the high-voltage conductor is made of copper, the insulating gas used is SF_6, and the spacer is made of epoxy resin. The operating conditions include a temperature of 313.15K, a voltage of 320kV, and a temperature gradient (ΔT) of 20K between the conductor and the enclosure, with the dimensions of GIL to be as shown in Fig. 2.

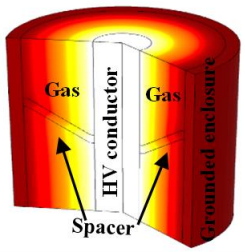


Fig. 1. Structure of GIL

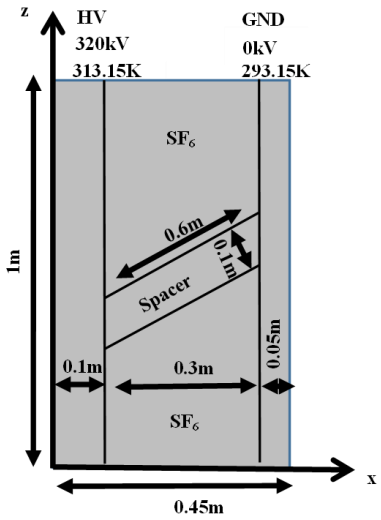


Fig. 2. 2D- axisymmetric geometry of HVDC GIL

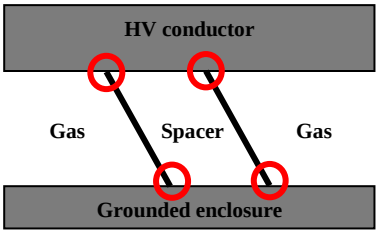


Fig. 3. Triple points of GIL

The presence of SF_6 gas holds the biggest benefit of the GIL as SF_6 has high dielectric strength compared to air [2] besides advantages like reliability, low electric loss, non-toxic, non-flammable etc. When high-voltage is given to the insulating materials (SF_6, spacer) there is an electric field stress that occurs. In GIL, electric filed stress can be seen at triple points as shown in Fig. 3. The electric field stress at the triple points should be reduced by applying field control techniques or else it can lead to system failure or partial discharge.

III. Approach

For time-varying electric field and negligible magnetic field, steady state equation can be generalised as shown in (1). Substituting Gauss law (2) into (1) results in (3), with current density in (4) and electric flux density in (5) for the linear materials we obtain (6), by using the electric field in (7), electro-quasistatic equation is determined in (8).

$$\mathrm{div}(\vec{J})+\frac{\partial}{\partial t}\rho = 0 \quad (1)$$

$$\mathrm{div}(\vec{D}) = \rho \quad (2)$$

$$\mathrm{div}(\vec{J})+\frac{\partial}{\partial t}\mathrm{div}(\vec{D}) = 0 \quad (3)$$

$$\vec{J} = \sigma\vec{E} \quad (4)$$

$$\vec{D} = \varepsilon_0\varepsilon_r\vec{E} = \varepsilon\vec{E} \quad (5)$$

$$\mathrm{div}(\sigma\vec{E})+\frac{\partial}{\partial t}\mathrm{div}(\varepsilon\vec{E}) = 0 \quad (6)$$

$$\vec{E} = \text{-grad}(\phi) \quad (7)$$

$$\mathbf{div}(\sigma\ \mathbf{grad}(\phi))+\frac{\partial}{\partial t}\mathbf{div}(\varepsilon\ \mathbf{grad}(\phi)\ ) = \mathbf{0} \quad (8)$$

where, ε is the permittivity with ε_r being the relative constant and the dielectric constant $\varepsilon_0 = 8{,}854 \cdot 10^{-12}$ V/m, σ is electric conductivity, $\vec{J}$ is current density, ρ is charge, $\vec{D}$ is electric flux density, ϕ is electric potential, $\vec{E}$ is electric field. Depending on the voltage state, the electric conductivity or permittivity is decisive for the electric field. Under the DC condition (resistive electric field), the electric conductivity is decisive [4] is given by:-

$$\mathrm{div}(\sigma\vec{E})=0 \quad (9)$$

Slow time variation of electro-quasistatic field applied to the insulating materials (SF6 gas and spacer) results in slow accumulation of charge within these materials [2]. Ability to conduct electric current in the insulating materials is limited, therefore when the electric field is applied, chargers are not able to flow through the material and it gets trapped within these materials, leading to an additional electric field. Changing the voltage rapidly results in transient capacitive electric field which is similar to AC condition, where permittivity is decisive is given by:-

$$\mathrm{div}(\varepsilon\vec{E})=0 \quad (10)$$

Simulations are performed in COMSOL Multiphysics software in stationary and transient fields, their respective effects are observed.

IV. Stationary Simulations

Under the DC steady state, 2D-axisymmetric geometry as shown in Fig. 2 is simulated assuming electric conductivity and permittivity of the spacer material and SF_6 gas as constant [1] and therefore, the electric field stress can be high at the triple points as shown in Fig. 4. to reduce the electric field stress at the triple points, field control techniques should be applied. There are several techniques that can be used to control the electric field stress. In this paper, two field control techniques are used i.e., Geometric Field Control and Field Graded Material. Under the geometric field control technique, changing the geometry of the spacer results in redistributed electric field with reduced electric filed stress.

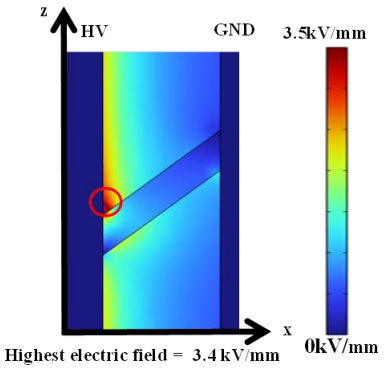


Fig.4. Stationary simulation of DC steady state

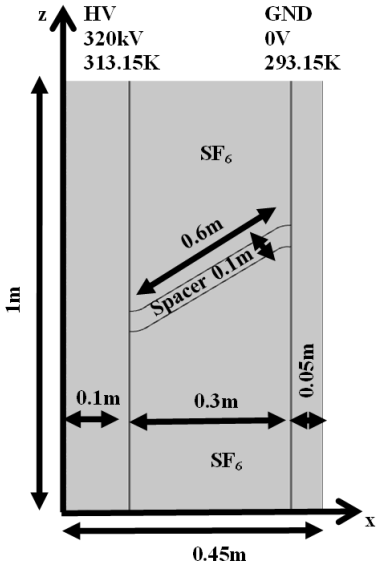


Fig. 5. 2D- axisymmetric geometry of HVDC GIL with cone-type spacer

Simulating 2D- axisymmetric geometry with cone-type spacer [2] as shown in Fig. 5, assuming electric conductivity and permittivity of the spacer material and SF_6 gas as constant (similar to the stationary simulation of DC steady state), results in reduction of electric field stress compared to DC

steady state mainly at the triple points as shown in Fig. 6. For the geometric field control technique, assumption of constant electric conductivity for the spacer material and SF6 gas is replaced with an equation of electric conductivity for the spacer material (11) and SF6 gas (12), where permittivity remains constant as previous simulations, it is possible to observe a shift in the electric field stress towards the other end of the triple point in the system, which is known as the field inversion effect as shown Fig. 7.

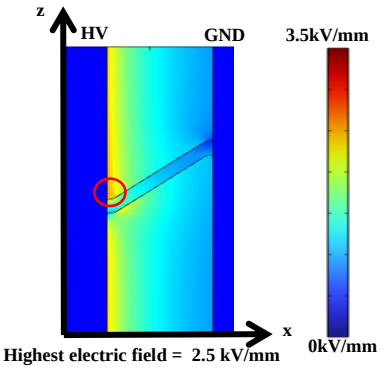


Fig. 6. Stationary simulation of geometric field control technique

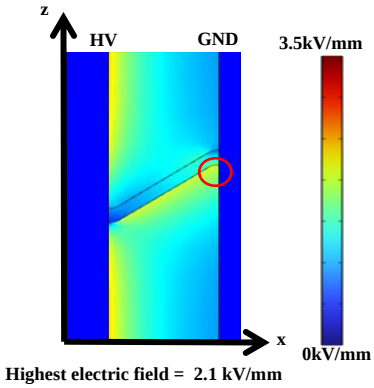


Fig. 7. Stationary simulation of electric conductivity model

Equation for electric conductivity for the spacer material is given by:-

$$\kappa_{spacer}=8.5 * e^{\left(\frac{-W}{k_B * T}\right)} * e^{a * \vec{E}} \quad (11)$$

where, k_B=8.617·10−5 eV/K is the Boltzmann constant, W=0.095 eV the activation energy, and a is constant [1, 2, 4], which is dependent on temperature T and electric field $\vec{E}$ [1, 2, 3, 4]. Equation for electric conductivity for the SF6 gas is given by:-

$$K_{SF6}=K * (\alpha+\beta\cdot(\gamma+|\vec{E}|/\vec{E}_x)^{\zeta}) * (1/(\varrho + \epsilon * |\vec{E}|/\vec{E}_y)^{l}) * e^{(\zeta * P)} * e^{(\nu * T)} \quad (12)$$

where K,α, β, γ, $\vec{E}_x$, ζ, ϱ, ϵ, $\vec{E}_x$, l, ν and ζ are constants. The geometric field control technique can be enhanced by incorporating an electric conductivity model for the spacer material and SF6 gas using equations (11) and (12), respectively, along with the use of FGM to distribute the electric conductivity around the spacer material as shown in the Fig. 8, with the equation for FGM Electric conductivity model for spacer material is given by:-

$$FGM_{conductivity}=\kappa_{spacer} * (1+((K_i -1) * e^{(-(r-r_i)/c)}) + ((K_a -1) * e^{((r-r_a)/q)})) \quad (13)$$

where, K_i, K_a, r_i, r_a, c, q, are the constants and r is the spatial variation of the spacer material, results in uniform distribution

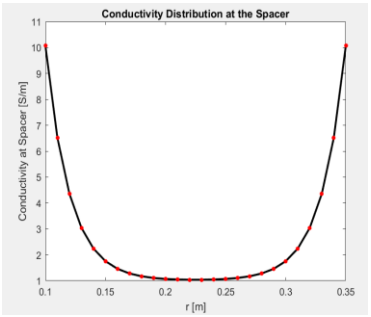


Fig. 8. Plot of electric conductivity distribution around spacer

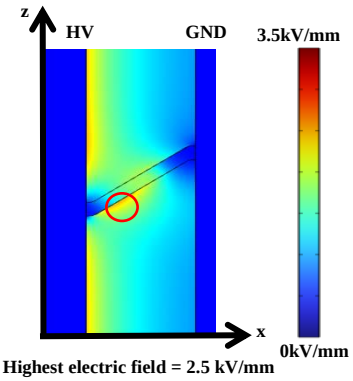


Fig.9. Stationary simulation of FGM

of the spacer material, results in uniform distribution of electric field around the spacer, with minimal electric field stress at the triple points compared with previous models, by increasing the electric field in the middle of the spacer as shown in Fig. 9. Hence, DC-FGM with electric conductivity distribution around the spacer gives better and uniformity in the results.

V. TRANSIENT SIMULATIONS

Under DC-on process the voltage level changes rapidly and results in transient capacitive electric field which is similar to AC condition, where permittivity is decisive [1, 2, 4, 5]. Therefore, under both DC & AC conditions, applying only electric conductivity model is not sufficient to obtain the better result, so the combination of electric conductivity and permittivity models are simulated further in this paper to obtain better results than the previous simulations. DC-on process is performed by applying DC voltage as given by:-

$$V0=320 * 10^3 * (-\exp^{(-70*t)}+1) \quad (14)$$

where t is time, to the conductor within the switching time of 0.01s after which the DC-steady state is reached and the saturation as shown in Fig. 10. Transient sismulation was applied for the DC-on process of the geometric field control technique (as previously analyzed in a stationary simulation), with a simulation time of $6e^5$(s) and time steps of 1000. The electric conductivity model in equations (11) and (12) was used for the insulating materials, with a constant permittivity for the insulating materials. As a result, reduced electric field stress is obtained as shown in Fig. 11 compared with the stationary simulation of electric conductivity model.

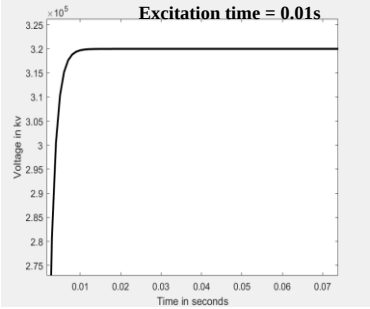


Fig. 10. Plot showing the switching time of the DC voltage

Transient simulation was applied for the DC-on process of the FGM (as previously analyzed in a stationary simulation) with similar parameters as in DC-on process geometric field control technique, but permittivity of the spacer is not constant rather it is distributed by the equation of permittivity around the spacer is given by:-

$$FGM_{permittivity} = 192 * r^2 - 86.4 * r + 10.72 \quad (15)$$

where r is the spatial variation of the spacer, and the distribution is shown in Fig. 12. Therefore the simulation results in further reduction of electric field stress, particularly at the triple points, and a more uniform electric field around the spacer, as demonstrated in Fig. 13. As a result, the DC-on process and transient simulation for the FGM provide the best results compared to the previous models investigated in this study.

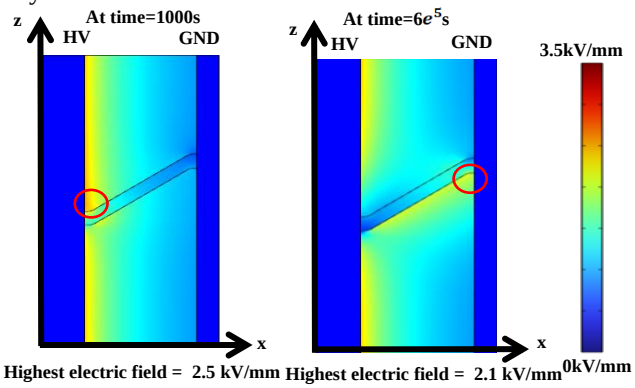


Fig. 11. Transient simulations of electric conductivity model

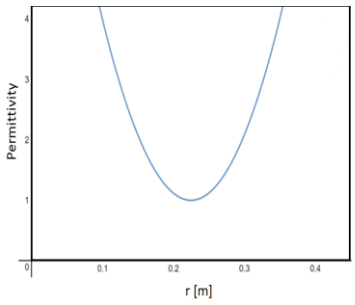


Fig. 12. Plot of permittivity distribution of spacer

VI. TEMPERATURE AND CHARGE DISTRIBUTION

Thermal stress occurs during operation and can lead to reduced system efficiency, insulation failure [6], etc. Fig. 14. shows 2D model of temperature distribution of GIL. The

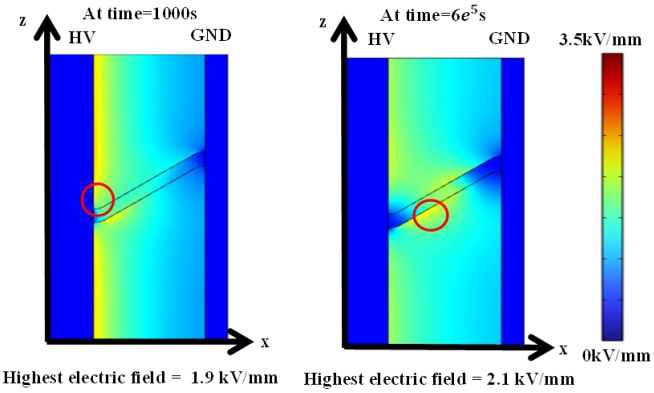


Fig. 13. Transient simulations of FGM

temperature is high near the conductor because electric current flows through the conductor in GIL and ohmic losses lead to a rising temperature.

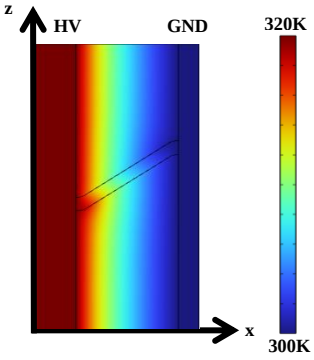


Fig. 14. 2D model of temperature distribution of GIL

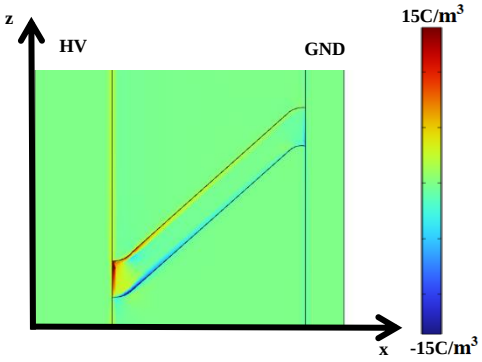


Fig. 15. 2D Charge distribution of geometric model

The charge distribution in GIL can affect the electric field distribution and in turn affect the performance and reliability of the system. When high voltage is applied to GIL, electric field in the insulating materials is created, which allows the charge carriers (electrons and ions) to move within the gas [7] therefore resulting in an accumulation of charge in the system mainly at triple points. This is the primary factor to determine the charge distribution. Next, properties of the insulating materials (electric conductivity and permittivity) can affect the charge distribution. Decay of charge can be observed when the electric conductivity and permittivity are assumed to be constant [8] which can be seen in Fig. 15. and reduced decay of charge can be observed in Fig. 16. where electric

conductivity model is applied for GIL where conductivity is not constant.

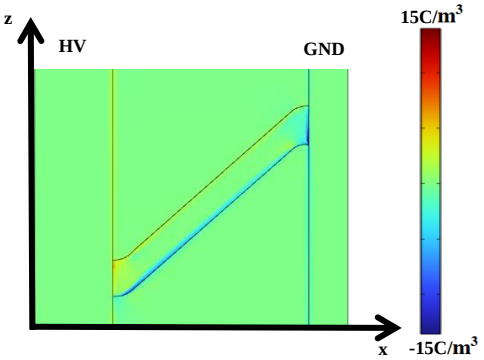


Fig. 16. 2D Charge distribution of electric conductivity model

VII. CONCLUSION

Different simulations were performed on GIL, where a combination of permittivity and electric conductivity distribution of FGM under DC-on process condition were investigated. FGM successfully reduced the electric field stress, especially at the triple points between gas, spacer and conductor/enclosure, giving uniform electric field around the spacer.

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